

MetroPulse — Executive Memo

Urban Mobility Intelligence | NYC | April 1–June 30, 2024

Executive Summary:

MetroPulse analyzed 10.59 million cleaned NYC taxi trips across April–June 2024, combining taxi, weather, zone, and subway data. Demand was highest in May, while the strongest recurring demand concentration occurred during the 5 PM–8 PM period, which represented 25.62% of all trips. Weather analysis also showed higher observed hourly taxi demand during rain hours, although the analysis is observational and does not establish that rain caused the increase.

Key Findings:-

- **Demand:** May recorded 3.66M trips, compared with 3.46M in April and 3.48M in June.
- **Peak period:** 5 PM–8 PM accounted for 2.71M trips (25.62%).
- **Geographic concentration:** Major demand and revenue concentrations were observed around Midtown, Upper East Side, and airport zones.
- **Weather:** Rain hours averaged approximately 5,524 trips/hour, compared with 4,725 trips/hour during no-rain hours. This is an association, not a causal estimate.
- **Data quality:** The final fact table reconciled exactly with the cleaned taxi dataset, and all 10 automated quality tests passed.
- **Fare stability:** April vs. May average charged amount differed by approximately \$0.90, with a 95% confidence interval of \$0.72–\$1.09.

Recommended 4-Week Pilot:

Initiative 1 — Peak-Hour Capacity Planning

Use the 5 PM–8 PM demand pattern to test more proactive capacity planning during peak hours.

- Baseline: 2.71M peak-period trips
- Scenario range: a 2%–5% planning scenario corresponds to approximately 54K–136K trips across the observed peak-period volume.
- Track: hourly demand, completed trips, median trip duration, and average charged amount.
- Risk: demand patterns may vary outside the study period.

Initiative 2 — Weather-Responsive Capacity Planning

Test whether weather-linked demand patterns can improve short-term capacity planning.

- Observed rain-hour demand: **5,524 trips/hour**
- Observed no-rain demand: **4,725 trips/hour**
- Scenario range: a 2%–5% planning scenario corresponds to approximately 38K–95K trips across the observed rain-hour volume.
- Track: hourly demand, trip completion, median duration, and weather conditions.
- Risk: the relationship is observational; weather should not be treated as the sole explanation for demand changes.

Decision Framework:

Run both initiatives for 4 weeks, compare results with the baseline, and evaluate changes using confidence intervals and sensitivity checks. Continue, modify, or stop each initiative based on observed operational metrics rather than assuming causal effects from historical correlations.

Data period: April 1–June 30, 2024

Sources: NYC TLC taxi data, NYC taxi-zone data, Open-Meteo weather data, and MTA subway ridership data.

Dashboard: Deployed MetroPulse application with interactive filters and data-quality monitoring.